

The Effect of Normality Assumption on Canonical Correlation Analysis of Overthinking Dimensions and Decision-Making Quality

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Abstract - This study discusses the effect of multivariate normality violations on Canonical Correlation Analysis in evaluating the relationship between overthinking and decision-making quality among college students. A quantitative survey design was conducted by distributing questionnaires to a sample of 390 undergraduate students at Bina Nusantara University Kemanggis. This study analyzed three dimensions of overthinking consisting of control, indecisiveness, and regret and five dimensions of decision-making quality consisting of speed, confidence, consistency, evaluation, and satisfaction. Assumption tests were conducted to test for linearity, multivariate normality, and multicollinearity. From the assumption tests, violations of multivariate normality were found. To address this, several transformation-based methods and kernel-based approaches were compared before conducting Canonical Correlation Analysis. The results showed that the transformation-based approach produced very stable canonical correlations with consistent significance patterns across methods. In contrast, the nonlinear kernel approach produced too high canonical correlations, indicating overfitting. Overall, the transformation-based approach provided more stable and easier-to-interpret results compared to the nonlinear kernel model.

Keywords: Canonical Correlation Analysis, Multivariate Normality, Overthinking, Decision Making

I. INTRODUCTION

Canonical Correlation Analysis (CCA) is a multivariate analysis method as a tool to measure and determine the correlation between two groups in a study [1]. CCA can be used for various research subjects, one of which is research that focuses on mental and psychological health topics because this research generally uses measurements with questionnaire-based data [2] [3]. There is previous research that shows that measurement data for human characteristics, personality, and behavior can violate the multivariate normal assumption which will later influence how to interpret canonical correlations [4].

The data must meet several assumptions, namely multivariate normality, linearity, homoscedasticity, and multicollinearity so that the CCA results can be interpreted properly [5]. Of the numerous assumptions that must be met for CCA, there is one assumption that plays an important role in guaranteeing accurate statistical inference, namely the multivariate normality assumption [6]. As is known, psychological data, for example regarding overthinking, often shows a skewed and uneven distribution across various populations, so this issue is very relevant to this topic [8]. To

measure the overthinking variable, several comparable dimensions are needed to support the measurement of the variable, so that later CCA can become an appropriate framework for examining the multivariate relationship of these dimensions.

To overcome multivariate normality violations, there are several approaches to overcome them before conducting CCA, these approaches are transformation-based approaches such as log transformation and Box-cox [9] [10]. Transformations can be used to reduce the impact of outliers by reducing large data variations [11]. Nevertheless, transformation procedures may alter the interpretability of psychometric variables and do not always successfully restore multivariate normality, especially when the original data contain complex nonlinear relationships [12]. Then for nonlinear problems, the kernel function method is adopted, the kernel function method is to map a low-dimensional space to a high-dimensional feature space [13].

Despite the increasing use of Kernel CCA and other non-parametric multivariate methods, limited studies have directly compared how different normality-handling strategies influence canonical correlation outcomes in psychometric research. Existing literature mainly emphasizes methodological performance rather than evaluating whether bypassing multivariate normality assumptions changes the magnitude and interpretation of canonical relationships. Consequently, insufficient evidence exists regarding the differences between transformation-based and kernel-based approaches when applied to multidimensional psychological constructs such as overthinking and decision-making quality.

Therefore, this study aims to examine the effect of multivariate normality violations on Canonical Correlation Analysis of overthinking dimensions and decision-making quality. Specifically, this study compares transformation-based approaches and Kernel Canonical Correlation Analysis to evaluate differences in canonical correlation estimates and interpretability under non-normal conditions. The study addresses the following research question: *How does the violation of multivariate normality affect CCA results and to what extent do different normality-handling strategies (transformation-based and kernel-based) produce divergent canonical correlations?*

The hypothesis in this study states that methods that ignore the assumption of multivariate normality will produce higher canonical correlation values compared to the transformation approach. Methodologically, this study contributes to a deeper understanding of how violations of multivariate normality affect the results of canonical correlation analysis in psychological research. Practically, this research is expected to help researchers choose more appropriate analytical techniques in interpreting psychometric data.

II. DATA AND METHODOLOGY

A. Dataset and Variables

This quantitative research used questions designed for distribution to students at Bina Nusantara University (BINUS) Kemanggisian. An online questionnaire using the Google Forms platform was distributed from April to May 2026 using a 6-point Likert scale calculation ranging from 1 (Strongly Disagree) to 6 (Strongly Agree) without any options leading to neutral statements to prevent respondents from looking for safe answers.

The sampling procedure applied simple random sampling to ensure fair representation across the target student population. The minimum sample size was calculated using the Slovin formula presented in Equation (1).

$$n = \frac{N}{1 + Ne^2} \quad (1)$$

where n represents the minimum sample size, N denotes the population size, and e is the margin of error.

In this study, the population consisted of active Bina Nusantara University students from cohorts B27, B28, and B29 in the 2026 academic year. The population included 3,675 students from cohort B27, 3,614 students from cohort B28, and 3,660 students from cohort B29, resulting in a total population of $N = 10,949$ students. A margin of error of 5% ($e = 0.05$) was applied. Therefore, the minimum required sample size for this study was 386 respondents.

The study investigated the relationship between two multivariate variable sets: Overthinking (X) and Decision-Making Quality (Y). The Overthinking construct consisted of five dimensions: Rumination, Control, Indecisiveness, Negative Thinking, and Regret. Meanwhile, the Decision-Making Quality construct consisted of five dimensions: Speed, Confidence, Consistency, Evaluation, and Satisfaction. Each dimension was measured using four questionnaire items.

Prior to Canonical Correlation Analysis (CCA), construct validity and reliability assessments were conducted to ensure measurement adequacy. Construct validity was evaluated using Confirmatory Factor Analysis (CFA), which is often used in psychological research when developing measurement models for psychological constructs [14]. Standardized loadings were used to assess the contribution of each indicator to its latent construct. Indicators were considered adequate when they had loading values of at least 0.50, with values above 0.70 considered strong [15]. A coefficient omega index estimates how reliably an observed composite score measures a target construct as represented by a factor in a factor-analysis model; as such, the accuracy of omega estimates is likely to

depend on correct model specification [16]. A minimum reliability threshold of $\omega > 0.70$ was applied based on recent psychometric reliability recommendations [17].

B. Assumption Testing and Data Preparation

Prior to Canonical Correlation Analysis (CCA), several preprocessing and assumption-testing procedures were conducted to ensure the suitability of the multivariate analysis framework. Since CCA relies on multivariate normality, linearity, homoscedasticity, and multicollinearity preliminary statistical diagnostics were performed before estimating canonical functions [5]. All preprocessing and statistical analyses were implemented in the R statistical environment using relevant multivariate and psychometric analysis packages.

The first assumption evaluated was linearity between the Overthinking variable set (X) and the Decision-Making Quality variable set (Y) the linearity assessment is important because classical CCA assumes that relationships between canonical variates can be adequately represented using linear combinations of variables [5]. To assess potential nonlinearity, each Overthinking dimension was regressed against the composite Decision-Making Quality score using the Ramsey Regression Equation Specification Error Test (RESET). Ramsey's RESET test can be used to determine the linear relationship between variables (linear or not) uses Ordinary Least Squares (OLS) to minimize the sum of the squared errors of each observation [18].

Multivariate normality was subsequently assessed because the statistical significance testing of canonical correlations, particularly Wilks' Lambda, depends on multivariate normality assumptions [19]. The study employed Mardia's multivariate skewness and kurtosis statistics, which are widely used to evaluate multivariate distributional adequacy in psychometric and multivariate statistical analysis [20]. Mardia's multivariate skewness statistic is defined in Equation 2:

$$b_{1,p} = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n [(x_i - \bar{x})^T S^{-1} (x_j - \bar{x})]^3 \quad (2)$$

while the multivariate kurtosis statistic is defined in Equation 3:

$$b_{2,p} = \frac{1}{n} \sum_{i=1}^n [(x_i - \bar{x})^T S^{-1} (x_i - \bar{x})]^2 \quad (3)$$

where n denotes the sample size, S denotes the covariance matrix among those dimensions, and x_i represents the vector of respondent scores across the Overthinking or Decision-Making Quality dimensions. The normality assessment was conducted separately for both variable groups before applying transformation-based and kernel-based procedures.

To evaluate potential multicollinearity among predictor dimensions, Variance Inflation Factor (VIF) and tolerance statistics were computed. Multicollinearity assessment is necessary because highly correlated predictor variables may destabilize covariance estimation and reduce interpretability in multivariate linear models. A VIF values below 10 and tolerance values above 0.10 were considered acceptable

indicators of non-problematic multicollinearity [21]. Multicollinearity diagnostics focused on the Overthinking dimensions because instability in predictor-side covariance structures primarily affects the estimation and interpretability of canonical weights in the X variable set.

C. Normality-Handling Strategies

The psychometric data used in this study often exhibit non-normal distributions, skewed, and bounded distributions, this study will compare several methods for handling multivariate normality for later use in Canonical Correlation Analysis (CCA). The approaches are divided into two categories: transformation-based normalization approaches and kernel-based distribution-free approaches. All methods will be used for all dimensions to ensure consistency throughout the analysis procedure.

1) Transformation-Based Methods

Transformation-based techniques were applied to reduce skewness, stabilize variance, and enhance approximation toward multivariate normality before performing classical CCA [9] [10]. Eight preprocessing approaches were evaluated: raw (untransformed baseline), logarithmic transformation, square-root transformation, Box-Cox transformation, Yeo-Johnson transformation, Nonparanormal (NPN) transformation, rank-based quantile normalization, and Ordered Quantile (ORQ) normalization.

The logarithmic and square-root transformations were applied as conventional variance-stabilizing procedures commonly used for positively bounded data [22] [23]. For power-based normalization, the Box-Cox transformation was implemented using Equation 4

$$y(\lambda) = \frac{y^\lambda - 1}{\lambda}, \lambda \neq 0 \quad (4)$$

where y represents the dimension-level psychometric scores obtained from the questionnaire responses and λ controls the degree of distributional adjustment applied to the scores [22]. Since psychometric dimension scores may exhibit heterogeneous distributional characteristics, the Yeo-Johnson transformation was additionally applied because it accommodates broader distributional conditions without requiring strictly positive observations [22]. The Yeo-Johnson transformation is defined in Equation 5:

$$y^{(\lambda)} = \begin{cases} \frac{[(y+1)^\lambda - 1]}{\lambda}, & y \geq 0, \lambda \neq 0 \\ \log(y + 1), & y \geq 0, \lambda = 0 \\ -\frac{[(-y+1)^{2-\lambda} - 1]}{2-\lambda}, & y < 0, \lambda \neq 2 \\ -\log(-y + 1), & y < 0, \lambda = 2 \end{cases} \quad (5)$$

To further address non-normality, semiparametric normalization procedures were also evaluated. The Nonparanormal (NPN) transformation estimates Gaussian-like marginals through rank-based semiparametric mapping [24], while rank-based quantile normalization and Ordered Quantile (ORQ) normalization transform observations according to empirical cumulative distribution structures [25] [26]. These approaches were included because rank-based

normalization methods are frequently applied in high-dimensional multivariate and psychometric preprocessing workflows.

After completing each stage of the transformation process, multivariate normality testing and canonical correlation analysis will be performed to ensure a fair comparison of each method.

2) Kernel-Based Methods

To overcome the problem of violation of the normality assumption, apart from using a transformation-based approach, the use of Kernel Canonical Correlation Analysis (KCCA) can be considered as an alternative to capture the nonlinear relationship between the variables of a set [27]. KCCA is able to project variables into the Reproducing Kernel Hilbert Space (RKHS) so that it allows for the estimation of canonical relationships without relying on the assumption of linearity or multivariate normality, which is very different from CCA [28].

Three kernel configurations were evaluated: linear kernel, Radial Basis Function (RBF), and polynomial. For a basis equivalent to CCA, a linear kernel with the function defined in Equation 6 can be utilized. However, if we wish to include an evaluation of potential nonlinear relationships in the measurement data of human characteristics and behavior, we can utilize the RBF and polynomial kernels [29]:

$$K(x_i, x_j) = x_i^T x_j \quad (6)$$

where x_i and x_j represent respondent-level psychometric feature vectors and $x_i^T x_j$ represents their inner product. The RBF kernel function is defined in Equation 7:

$$K(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right) \quad (7)$$

where σ represents the kernel bandwidth parameter [30]. The RBF bandwidth parameter ($\sigma = 0.1$) was selected after a grid search over $\sigma \in 0.01, 0.05, 0.10, 0.50, 1.00$. The test results show that the $r_1^* > 0.99$ indicates overfitting and is not specific to a particular σ value, but rather an inherent characteristic of the applied non-linear kernel method. Meanwhile, the polynomial kernel is defined in Equation 8:

$$K(x_i, x_j) = (x_i^T x_j + c)^d \quad (8)$$

where c is a constant term and d denotes the polynomial degree [29]. The reason for choosing a polynomial kernel degree ($d = 2$) is to capture nonlinear relationships while maintaining model interpretability and computational stability.

All processes, from preprocessing to modeling, were implemented in R using the multivariate analysis package. All transformation-based and kernel-based methods were evaluated using a defined set of variables and evaluation criteria to ensure methodological consistency.

D. Canonical Correlation Analysis Procedure

Canonical Correlation Analysis (CCA) was employed to examine the multivariate relationship between the Overthinking variable set (X) and the Decision-Making Quality variable set (Y). CCA identifies linear combinations of variables from both sets that maximize the correlation between the two multivariate constructs [31]. The canonical correlation optimization problem is defined in Equation 9 and 10:

$$\rho = \max_{a,b} \frac{a^T \Sigma_{XY} b}{\sqrt{a^T \Sigma_{XX} a} \sqrt{b^T \Sigma_{YY} b}} \quad (9)$$

$$\rho = \max_{a,b} \text{corr}(a^T X, b^T Y) \quad (10)$$

where Σ_{XX} and Σ_{YY} denote the covariance matrices of the Overthinking and Decision-Making Quality variable sets, respectively, while Σ_{XY} represents the cross-covariance matrix between both variable groups. The vectors a and b represent the canonical coefficient vectors (weights) used to construct the linear combinations of the Overthinking dimensions and Decision-Making Quality dimensions. The objective of CCA is to identify weighted combinations of both psychological constructs that maximize the correlation between the resulting canonical variates.

The canonical variates are subsequently constructed in Equation 11 and 12:

$$U = a^T X \quad (11)$$

$$V = b^T Y \quad (12)$$

where U represents the latent canonical variate summarizing the Overthinking dimensions, while V represents the latent canonical variate summarizing Decision-Making Quality dimensions. The resulting canonical correlations measure the maximum association between linear combinations of the two multivariate constructs [32].

To evaluate the statistical significance of the canonical functions, Wilks' Lambda (Λ) significance testing was applied for all transformation-based CCA procedures [33]. Wilks' Lambda evaluates whether the canonical correlations collectively explain a statistically significant proportion of shared variance between the two variable sets. Smaller values of Wilks' Lambda indicate stronger multivariate associations between canonical variates [33].

In addition to canonical correlations, canonical loadings (structure coefficients) were examined to evaluate the contribution of each original variable to its corresponding canonical variate [34]. Canonical loadings are particularly important in psychometric multivariate analysis because they provide substantive interpretation regarding which dimensions contribute most strongly to the canonical relationship structure. Following common multivariate interpretation guidelines, loadings with absolute values greater than or equal to 0.50 were considered substantively meaningful [35].

To ensure methodological consistency, all transformation-based and kernel-based approaches were evaluated using

identical variable sets, preprocessing procedures, and interpretation criteria. Comparative evaluation focused on the magnitude of canonical correlations, significance structure, and interpretability of canonical loadings across all normality-handling strategies.

III. RESULTS AND DISCUSSION

A. Data Preprocessing and Dimension Selection

Before running any multivariate analysis, a series of preprocessing steps was applied to ensure only valid and reliable dimensions entered the canonical correlation model. Validity was assessed via Confirmatory Factor Analysis (CFA), where a dimension was considered valid only if all its items had standardized factor loadings exceeding 0.7 and were statistically significant ($p < 0.05$) [14]. Eight out of ten dimensions passed this criterion. The two excluded dimensions were Rumination and Negative Thinking: one item in Rumination (X_4 , loading = 0.648) and one item in Negative Thinking (X_{13} , loading = 0.685) fell below the threshold, causing both dimensions to be dropped. The overall CFA model showed acceptable fit ($CFI = 0.898$, $TLI = 0.886$, $RMSEA = 0.066$, $SRMR = 0.057$) [36].

To ensure that the data is reliable, Reliability is evaluated using McDonald's omega using a threshold of $\omega > 0.70$ [37]. Of all the dimensions that were checked, all of these dimensions successfully passed the McDonald omega standard limit ranging from 0.843 (Rumination) to 0.901 (Confidence) which can be interpreted that all dimensions have strong internal consistency across all aspects. Since Rumination and Negative Thinking had already been excluded on validity grounds, the final analysis used three Overthinking dimensions, Control ($\omega = 0.859$), Indecisiveness ($\omega = 0.869$), and Regret ($\omega = 0.887$) and five Decision-Making Quality dimensions, Speed, Confidence, Consistency, Evaluation, and Satisfaction. Dimension-level scores were computed as the mean of each dimension's items.

Table I presents the descriptive statistics for the eight selected dimensions. Overthinking dimensions showed higher mean scores (range: 4.23–4.51) compared to Decision-Making Quality dimensions (range: 3.77–4.34). All distributions were negatively skewed, consistent with the bounded nature of six-point Likert scales where respondents tend to cluster above the midpoint.

TABLE I. DESCRIPTIVE STATISTICS OF FINAL DIMENSION SCORES

Dimension	n	Mean	SD	Min	Max	Skewness
Group X: Overthinking Dimensions						
Control	390	4.435	1.031	1.50	6.00	-0.823
Indecisiveness	390	4.513	1.011	1.25	6.00	-0.916
Regret	390	4.231	1.131	1.00	6.00	-0.776
Group Y: Decision-Making Quality Dimensions						
Speed	390	3.769	0.887	1.00	6.00	-0.143
Confidence	390	3.931	0.914	1.00	6.00	-0.193
Consistency	390	3.896	0.905	1.00	6.00	-0.194
Evaluation	390	4.341	0.923	1.75	6.00	-0.165
Satisfaction	390	4.088	0.877	1.50	6.00	-0.174

Note: All dimensions show negative skewness, consistent with the bounded nature of six-point Likert scales.

B. Assumption Testing

1) Linearity

The Ramsey RESET test was applied to each Overthinking dimension against the composite Y score (mean of all five Decision-Making Quality dimensions). As shown in Table II, all three dimensions passed the linearity test (all $p > 0.05$), confirming that the linear framework assumed by CCA is appropriate for this data [18].

TABLE II. RAMSEY RESET TEST FOR LINEARITY

Dimension (X)	F-stat	p-value	Result
Control	2.322	0.0995	Linear
Indecisiveness	1.068	0.3447	Linear
Regret	1.972	0.1405	Linear

2) Multivariate Normality

To test for multivariate normality, the Mardia test was applied to examine skewness and kurtosis on two groups of variables separately. After conducting the tests, as shown in Table III, both variables failed the normality test. This result is due to data using a Likert scale rarely meeting the assumption of multivariate normality [38]. The violation of the normality assumption prompted a comparison of several handling strategies as explained in the next section.

TABLE III. MARDIA'S MULTIVARIATE NORMALITY TEST RESULTS

Group	Test	Statistic	p-value	Result
Overthinking (X)	Mardia's Skewness	154.200	< 0.001	Not Normal
	Mardia's Kurtosis	11.122	< 0.001	Not Normal
Decision-Making Quality (Y)	Mardia's Skewness	335.639	< 0.001	Not Normal
	Mardia's Kurtosis	18.741	< 0.001	Not Normal

3) Multicollinearity

To determine whether there is multicollinearity, the Variance Inflation Factor (VIF) is calculated for the third dimension of overthinking. All VIF values were below 10 (range: 2.05 – 2.35) and all tolerance values exceeded 0.10 (range: 0.43 – 0.49), indicating no problematic multicollinearity [39]. The correlations among X dimensions ranged from 0.647 to 0.701, moderate but not extreme. Table IV summarizes the multicollinearity diagnostics.

TABLE IV. VIF AND TOLERANCE FOR OVERTHINKING DIMENSIONS

Dimension	VIF	Tolerance	Result
Control	2.051	0.487	No Multicollinearity
Indecisiveness	2.210	0.453	No Multicollinearity

Regret	2.345	0.426	No Multicollinearity
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C. Effect of Normality Handling on CCA Results

After checking the multivariate normality assumption, it was found that both groups of variables violated the multivariate normality assumption. This study compared the use of eleven methods to understand the assumptions of these eight approaches, using transformation-based and three kernel selections. The main question that arises is whether the choice of normality strategy significantly changes the results of the canonical correlation.

1) Transformation-Based Methods

Eight transformations were tested on the data: raw (no transformation, just used as a baseline), logarithmic, square root, Box-Cox, Yeo-Johnson, Nonparanormal (NPN) [40], rank-based with quantile normalization, and Ordered Quantile (ORQ) normalization [13].

After applying the transformation to the data, the assumption of normality was checked again using Mardia's test and CCA was run on the transformed data and Wilks' Lambda was used to test for significance. The results are shown in Table V.

A striking finding is how little the choice of transformation actually mattered. Across all eight methods, the first canonical correlation (r_1^*) ranged narrowly from 0.379 (Log) to 0.402 (Raw), a spread of less than 0.025. Every method produced all three canonical functions as statistically significant (all $p < .001$ for CF1 and CF2). Regarding normality improvement, Yeo-Johnson, NPN, Rank with qnorm, and ORQ all achieved partial normalization of the X group skewness (X skew p ranging from 0.053 to 0.339), though Y group normality remained unachieved across all methods, again consistent with the bounded structure of Likert data.

2) Kernel-Based Methods

Three kernel configurations were evaluated using Kernel CCA (kCCA): a linear kernel (equivalent to standard CCA), an RBF kernel ($\sigma = 0.1$), and a polynomial kernel (degree = 2). kCCA operates in a Reproducing Kernel Hilbert Space (RKHS) and does not require any distributional assumption [28], making it theoretically attractive for non-normal data. However, it is important to note that kCCA's primary purpose is capturing non-linear associations, not correcting distributional violations.

The kCCA linear kernel produced $r_1^* = 0.4023$, nearly identical to the standard CCA result, confirming internal consistency. The RBF kernel ($r_1^* = 0.9986$) and polynomial kernel ($r_1^* = 0.9997$) both produced near-unity canonical correlations, a clear sign of overfitting. The RBF kernel showed the same value for r_1^* and r_2^* (both 0.9986), which is a known numerical artifact of Cholesky decomposition in kCCA when the sample to dimension ratio is moderate [28]. These inflated results are not credible for individual-level psychometric data where true canonical correlations rarely approach 1.0.

TABLE V. CANONICAL CORRELATIONS ACROSS ALL METHODS

Method	r_1^*	r_2^*	r_3^*	p_{CF1}	CF sig.	Status
Transformation-Based Methods						
1. Raw (baseline)	0.4024	0.3146	0.1779	< 0.001	3	OK
2. Log	0.3789	0.2879	0.1860	< 0.001	3	OK
3. Sqrt	0.3951	0.3076	0.1863	< 0.001	3	OK
4. Box-Cox	0.3989	0.3038	0.1601	< 0.001	3	OK
5. Yeo-Johnson	0.3961	0.3031	0.1593	< 0.001	3	OK
6. NPN	0.3978	0.2897	0.1537	< 0.001	3	OK
7. Rank+qnorm	0.3950	0.2930	0.1644	< 0.001	3	OK
8. ORQ	0.3948	0.2920	0.1677	< 0.001	3	OK
Kernel-Based Methods (Distribution-Free)						
9. kCCA Linear	0.4023	0.4023	0.3144	N/A	N/A	OK
10. kCCA RBF ($\sigma = 0.1$)	0.9986	0.9986	0.6726	N/A	N/A	OVERFIT
11. kCCA Poly ($d = 2$)	0.9997	0.9997	0.5255	N/A	N/A	OVERFIT

Note: r^* = canonical correlation. N/A = Wilks' Lambda not available for kernel methods. OVERFIT = $r > 0.95$, not credible for psychometric Likert data.

Table V shows that CCA results using transformations tend to be stable regardless of the transformation used. The canonical correlation structure is robust to distributional adjustments. In contrast, nonlinear kernel methods such as RBF and polynomial methods produce very high estimates.

The results of all transformation methods showed similarities, with a range of 0.379 – 0.402, with differences of less than 0.025. The CCA model was ultimately chosen because the application of the transformation did not have a significant impact, but instead made the analysis more complex. This choice also aligns with the observed stability of CCA results when the sample size is sufficient [1].

D. Canonical Correlation Analysis Result

Significance tests for the three canonical functions of the selected model are presented in Table VI. The results show that the three functions are significant, as seen through the p-value, indicating that the relationship between overthinking and decision-making quality is multidimensional, meaning it cannot be combined into a single axis [33].

TABLE VI. WILKS' LAMBDA SIGNIFICANCE TESTS FOR CANONICAL FUNCTIONS

CF	Canonical r	Wilks' λ	F	df_1	p-value
1	0.4024	0.7312	8.446	15	< 0.001
2	0.3146	0.8725	6.757	8	< 0.001
3	0.1779	0.9684	4.183	3	0.006

The canonical loadings, correlations between original variables and their respective canonical variates are shown in Table VII. For CF_1 , all three Overthinking dimensions loaded positively and substantially ($|loading| \geq 0.50$ is the standard threshold for substantive interpretation [35]): Indecisiveness at 0.9910, Regret at 0.7759, and Control at 0.6268. This pattern suggests that CF_1 for X primarily captures a general tendency to overthink with indecisiveness as the strongest contributor.

TABLE VII. CANONICAL LOADINGS - RAW CCA ($|LOADING| \geq 0.50$ IN CF_1)

Variable	CF_1	CF_2	CF_3
Group X: Overthinking Dimensions			
Control	0.6268	-0.6060	-0.4897
Indecisiveness	0.9910	0.0561	-0.1217
Regret	0.7759	-0.5527	0.3043
Group Y: Decision-Making Quality Dimensions			
Speed	-0.8542	-0.4549	0.0397
Confidence	-0.9625	0.1271	-0.2187
Consistency	-0.7110	-0.0026	-0.5654
Evaluation	-0.6753	0.4326	0.0818
Satisfaction	-0.7586	0.3058	-0.3100

Canonical weights, which show how much each variable uniquely contributes to the canonical variate after accounting for correlations within the same set, were also looked at. For CF_1 , Indecisiveness had the largest weight among the OT variables (0.914), followed by Regret (0.203) and Control (-0.101). On the DMQ side, Confidence (-0.839) and Speed (-0.360) carried the biggest weights. The fact that the loadings and weights pointed in the same direction for CF_1 is a good sign for the stability of the solution, though interpretation was mainly based on loadings since the OT dimensions were moderately correlated with each other.

On the DMQ side, all five dimensions showed substantial negative loadings on CF_1 : Confidence, Speed, Satisfaction, Consistency, and Evaluation all came in below the threshold. The consistent negative direction means that the DMQ canonical variate moves opposite to the OT one, so higher Overthinking scores go hand in hand with lower Decision-Making Quality scores across the board.

CF_2 picked up a secondary pattern where Control and Regret loaded negatively while Evaluation and Satisfaction showed positive loadings on the DMQ side. This suggests that feeling regret and losing control over one's thoughts is specifically tied to lower evaluative confidence and reduced satisfaction with decisions, separate from the general overthinking tendency that CF_1 already captured. CF_3 , while statistically significant, mostly picks up leftover fine-grained variation and does not add much beyond what CF_1 and CF_2 already explain.

The main methodological takeaway from this study is that the choice of normality-handling strategy does not really change the CCA results as long as a transformation-based approach is used. The canonical correlation values stayed within a narrow band across all eight methods, and the significance pattern was identical every time. This

consistency strongly suggests that the association found in the data is a genuine pattern rather than something that just happened because of how the analysis was set up.

The kernel CCA results, however, tell a cautionary tale. The RBF and polynomial kernels both produced canonical correlations close to 1.0, which are clearly not plausible for individual-level psychometric data. These results are better understood as overfitting artifacts than genuine associations. Substantively, the CCA results point to a meaningful inverse relationship between overthinking dimensions and decision-making quality. Indecisiveness was by far the strongest predictor in the Overthinking set $loading = 0.991$ on CF1, which makes intuitive sense, difficulty committing to a decision is a direct behavioural manifestation of overthinking that immediately impairs decision outcomes. Regret 0.776 and Control 0.627 also contributed substantially, reflecting how anticipated post-decision regret and difficulty managing thought processes compound the overall impairment. On the quality side, Confidence took the largest hit $loading = -0.963$ followed by Speed -0.854 suggesting that overthinking most acutely undermines how certain people feel about their choices and how quickly they can make them.

This study has several limitations that need to be acknowledged. First, the variables were measured using Likert-type scales, which are generally ordinal and have a limited range. Consequently, multivariate normality could not be achieved regardless of the transformation applied. Second, because the sample was limited to students from specific classes, campuses, and regions, the findings could be limited to broader or more diverse groups. Third, despite exploring kernel CCA as a distribution-free alternative, the RBF and polynomial kernels produced significantly inflated canonical correlations due to overfitting, as the data is known to be difficult to calibrate without cross-validation. Future studies are expected to consider other CCA approaches that can address this issue more rigorously.

IV. CONCLUSION

From the research conducted, it can be concluded that one and the other groups of variables violate multivariate normality based on the standards of Mardia's skewness and kurtosis test. This means that psychological testing likert-scale data rarely meet the assumptions of classical distribution. One technique used is the use of transformation, a transformation-based approach that produces very consistent and stable canonical correlation outcomes. Among the eight transformation methods, the first canonical correlation remained within a narrow range (0.379 - 0.402), and all canonical functions showed an equivalent significance structure. This indicates that minor violations of multivariate normality do not substantially distort the overall interpretation of CCA when the sample size is adequate.

However, when we used additional approach, the kernel-based approach, the results were significantly different. The outcome from the linear kernel produced correlations similar to those of the classical CCA, the RBF and polynomial kernels, where the canonical correlation values were close to perfect, interpreted as indicating severe overfitting in the

model. These results indicate that abandoning the normality assumption can increase canonical correlations and reduce interpretability in low-dimensional psychometric datasets.

This study confirms a multidimensional negative relationship between overthinking and decision-making quality, where higher explanatory variables of overthinking such as indecisiveness, regret, and lack of control can be associated with explanatory variables of decision-making quality such as lower self-confidence, slower decision-making, and reduced satisfaction. Overall, this study concludes that transformation-based CCA remains more robust, easy to interpret, and appropriate than the nonlinear kernel-based approach for data with violations of normality.

REFERENCES

- [1] I. Akour, A. A. Rahamneh, B. Al Kurdi, A. Alhamad, I. Al-Makhari, M. Alshurideh and S. Al-Hawary, "Using the Canonical Correlation Analysis Method to Study Students' Levels in Face-to-Face and Online Education in Jordan," *Information Science Letters*, vol. 12, no. 2, pp. 901-910, 2023.
- [2] L. Li, P. Wang, S. Li, Q. Liu, F. Yu, Z. Guo, S. Jia and X. Wang, "Canonical correlation analysis of depression and anxiety symptoms among college students and their relationship with physical activity," *Scientific Reports*, vol. 13, pp. 1-10, 2023.
- [3] T. Zhong, "Physical activity motivations and psychological well-being among university students: a canonical correlation analysis," *Cognitive Neuroscience and Neuroimaging*, vol. 12, pp. 1-7, 2024.
- [4] H. Ferr, "The normal distribution is not normal in psychological data: Moving beyond parametric dogma," *PLOS Ment Health*, vol. 2, no. 8, pp. 1-4, 2025.
- [5] A. Temurtas, "The Use of Canonical Correlation Analysis in Educational Research: A Systematic Review of Turkish Context," *EDUCATIONE*, vol. 4, no. 2, pp. 32-60, 2025.
- [6] Z. Feng, M. Zhao and X. Qi, "Multivariate normality test based on Shannon's entropy," *Statistics*, vol. 59, no. 5, p. 1233-1252, 2025.
- [7] A. B. Downey, *Probably Overthinking It: How to use data to answer questions, avoid statistical traps, and make a better decisions*, vol. 212, Chicago: The University of Chicago Press, 2023.
- [8] B. Damsa, "Academic Overthinking: When Students Put Too Much Pressure on Success," *Journal Educational Verkenning*, vol. 7, no. 1, pp. 1-8, 2026.
- [9] A. Ramdhanian and N. A. K. Rifai, "Diagram Kendali Multivariat T2 Hotelling pada Instalasi Pengolahan Air Limbah," *Bandung Conference Series*, vol. 2, no. 2, pp. 189-196, 2022.
- [10] S. O. Sihombing, *Pengantar Metode Analisis Multivariat*, vol. 6, Pekalongan: PT Nasya Expanding Management, 2022, pp. 163-175.
- [11] Elisawati, A. Impron, A. Latif, L. Faizal, J. Budiasto, J. S. Pasaribu, A. N. Weking, A. J. Wahidin, Ismail, M. F. Almaliki, Isnawaty, A.

- Wijaya and R. Dewi, Machine Learning: Konsep, Implementasi, dan Aplikasi, vol. 8, Padang: Penerbit Get Press Indonesia, 2025, pp. 1-7.
- [12] G. Hatem, J. Zeidan, M. Goossens and C. Moreira, "Normality Testing Methods and The Importance of Skewness and Kurtosis in Statistical Analysis," *BAU Journal - Science and Technology*, vol. 3, no. 2, pp. 1-5, 2022.
- [13] Q. Chen and Y. Wang, "Key-performance-indicator-related state monitoring based on kernel canonical correlation analysis,," *Control Engineering Practice*, vol. 107, 2021.
- [14] D. Goretzko, K. Siemund and P. Sterner, "Evaluating Model Fit of Measurement Models in Confirmatory Factor Analysis," *Educ Psychol Meas*, vol. 84, no. 1, pp. 123-144, 2024.
- [15] N. N. S. P. Verawati, J. Rokhmat, A. Harjono and M. Makhrus, "Construct Validation of a Critical Thinking Disposition Instrument for Physics Education Students: An Exploratory and Confirmatory Factor Analysis Approach," *Lensa: Jurnal Kependidikan Fisika*, vol. 14, no. 1, pp. 1-21, 2026.
- [16] S. M. Bell, R. P. Chalmers and D. B. Flora, "The Impact of Measurement Model Misspecification on Coefficient Omega Estimates of Composite Reliability," *Educational and Psychological Measurement*, vol. 84, no. 1, pp. 5-39, 2024.
- [17] K. Stensen and S. Lydersen, "Internal consistency from alpha to omega?," *Tidsskr Nor Lægeforen*, vol. 142, pp. 1-3, 2022.
- [18] F. H. Junianto, A. A. R. Fernandes and R. Solimun, "Development Of Semiparametric Path Analysis Modeling Truncated Spline: Determinants Of Increased Regional Economic Growth," *BAREKENG: Journal of Mathematics and Its Applications*, vol. 18, no. 2, pp. 949-960, 2024.
- [19] E. Gonzalez-Estrada, J. A. Villasenor and R. Acosta-Pech, "Shapiro-Wilk Test for Multivariate Skew-Normality," *Computational Statistics*, vol. 37, no. 4, pp. 1985-2001, 2022.
- [20] U. A. Victor-Edema, "Comparative Analysis of Some Approaches to Multivariate Normality Test," *Faculty of Natural and Applied Sciences Journal of Scientific Innovation*, vol. 4, no. 2, pp. 154-164, 2023.
- [21] M. Nazligül, A. Yilmaz and M. Griffiths, "aming Addiction and Exercise Addiction: To What Extent Are They The Same or Different In Terms of Emotional Abuse and/or Emotional Neglect Etiologies?," *Int J Ment Health Addiction* 21, vol. 21, pp. 145-164, 2023.
- [22] Z. Cebeci, F. Ceritoglu, M. C. Guney and A. Unalan, "A Moment-Targeting Normality Transformation Based on Simultaneous Optimization of Tukey g-h Distribution Parameters," *Symmetry*, vol. 18, no. 3, p. 458, 2026.
- [23] R. M. West, "Best Practice in Statistics: The Use of Log Transformation," *Annals of Clinical Biochemistry: International Journal of Laboratory Medicine*, vol. 59, no. 3, pp. 162-165, 2022.
- [24] S. M. Foltz, C. S. Greene and J. N. Taroni, "Cross-platform normalization enables machine learning model training on microarray and RNA-seq data simultaneously," *Communications Biology*, vol. 6, no. 1, p. 222, 2023.
- [25] Y. Sun and X. He, "From Regression Rank Scores to Robust Inference for Censored Quantile Regression," *Canadian Journal of Statistics*, vol. 51, no. 4, pp. 1126-1149, 2023.
- [26] R. A. Peterson, "Finding Optimal Normalizing Transformations via bestNormalize," *The R Journal*, vol. 17, 2021.
- [27] Q. Chen and Y. Wang, "Key Performance Indicator Related State Monitoring Based on Kernel Canonical Correlation Analysis," *Control Engineering Practice*, vol. 107, 2021.
- [28] E. Ullah and R. Arora, "Generalization bounds for kernel canonical correlation analysis," *Transactions on Machine Learning Research*, 2023.
- [29] A. F. Rochim, K. Widyaningrum and D. Eridani, "Performance comparison of support vector machine kernel functions in classifying COVID-19 sentiment," *International Seminar on Research of Information Technology and Intelligent Systems*, p. 224, 2021.
- [30] A. Matropietro, C. Feldmann and J. Bajorath, "Calculation of Exact Shapley Values for Explaining Support Vector Machine Models Using the Radial Basis Function Kernel," *Scientific Reports*, vol. 13, no. 1, p. 19561, 2023.
- [31] J. A. C. Vargas, V. D. S. B. Gomes, R. Mezzomo and R. P. Maciel, "Multivariate Relationship between Major Constituents and Casein Fractions in Buffalo Milk using Canonical Correlation Analysis," *International Dairy Journal*, vol. 142, p. 105651, 2023.
- [32] A. Bykhovskaya and V. Gorin, "Canonical Correlation Analysis," *arXiv preprint arXiv*, 2024.
- [33] R. Cruz-Cano, A. Cohen and E. Mead-Morse, "Canonical Correlation Analysis of Survey Data: The SurveyCC R Package," *The R Journal*, vol. 16, no. 4, 2024.
- [34] A. R. McIntosh, "Comparison of Canonical Correlation and Partial Least Squares Analysis of Simulated and Empirical Data," *arXiv preprint*, 2021.
- [35] J. Smoluk-Sikorska and M. Malinowski, "An Attempt to Apply Canonical Analysis to Investigate the Dependencies between the Level of Organic Farming Development in Poland and the Chosen Environmental Determinants," *Energies*, vol. 14, no. 24, p. 8390, 2021.
- [36] A. Khademi, C. S. Wells, M. E. Oliveri and E. Villalonga-Olives, "Examining appropriacy of CFI and TLI cutoff value in multiple-group CFA test of measurement invariance to enhance accuracy of test score interpretation," *SAGE Open*, vol. 13, no. 4, 2023.
- [37] F. Orcan, "A comparison of Cronbach's alpha and McDonald's omega for ordinal data: Are they different?," *Int. J. Assess. Tools Educ.*, vol. 10, no. 4, pp. 709-722, 2023.
- [38] C. DiStefano, D. Shi and G. B. Morgan, "Collapsing categories is often more advantageous than modeling sparse data: Investigations in

the CFA framework," *Struct. Equ. Modeling*, vol. 28, no. 2, pp. 237-249, 2021.

- [39] C. C. Jeng, "Why a variance inflation factor of 10 is not an ideal cutoff for multicollinearity diagnostics," *J. Educ. Res. Dev.*, vol. 57, no. 2, pp. 67-92, 2023.

- [40] S. Na, M. Kolar and O. Koyejo, "Estimating differential latent variable graphical models with applications to brain connectivity," *Biometrika*, vol. 108, no. 2, pp. 425-442, 2021.